Pre-Lab 4: Mesh Analysis Methods

A. I_2 in terms of $I_1,\ R_2,\ R_3$. I_1 gets absorbed in when solving for I_2. The hand calculations are in the next page.

$$I_2 = \frac{\left(10\frac{R_3}{R_2+R_3} - 5\right)}{\left(-\frac{R_3^2}{R_2+R_3} + (R_2+R_3)\right)} = -375\,\mu\text{A}$$

B. Current values for I_1 and I_2

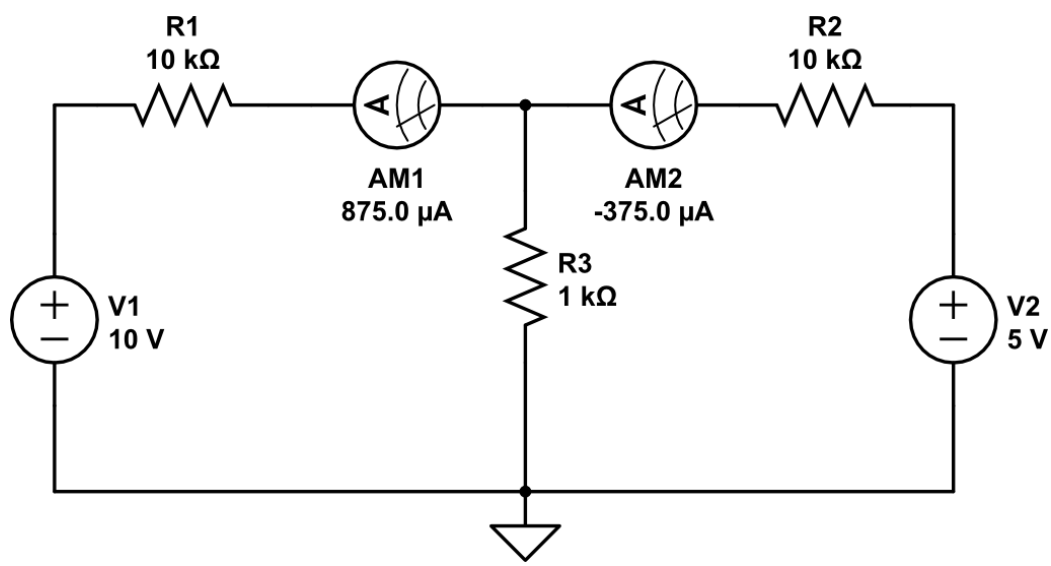
$$I_1 = \frac{I_2 R_3 + 10}{R_2 + R_3} = 875\,\mu\text{A}$$

Matrix operations can also be used.

$\mathcal{L}et$ col 1 be I_1 and col 2 be I_2, solving the system equations using matrix operations:

$$\begin{bmatrix} (R_1+R_3) & -R_3 & 10 \\ -R_3 & (R_2+R_3) & \text{-5} \end{bmatrix} = \begin{bmatrix} 11\,\text{k}\Omega & -1\,\text{k}\Omega & 10 \\ -1\,\text{k}\Omega & 11\,\text{k}\Omega & \text{-5} \end{bmatrix} \overset{\text{rref}}{\Longrightarrow} \begin{bmatrix} 1 & 0 & 875\,\mu\text{A} \\ 0 & 1 & -375\,\mu\text{A} \end{bmatrix}$$

C. Circuit Lab simulations



Part 1: Mesh Analysis

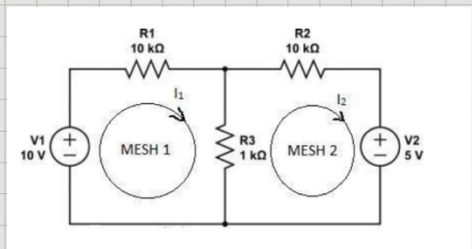


Figure 4.1

Loop 1 KVL:

$R_2 = R_1$

$$-10 + R_1(I_1) + R_3(I_1 - I_2) = 0$$
$$I_1(R_1 + R_3) - I_2 R_3 = 10$$

Loop 2 KVL:

$$R_3(I_2 - I_1) + R_2(I_2) + 5 = 0$$
$$-I_1 R_3 + I_2(R_2 + R_3) = -5$$

$$I_1 = \frac{I_2 R_3 + 10}{R_2 + R_3} \Rightarrow -\left[\frac{I_2 R_3 + 10}{R_2 + R_3}\right] R_3 + I_2(R_2 + R_3) = -5$$

$$\frac{-I_2 R_3^2}{R_2 + R_3} - \frac{10 R_3}{R_2 + R_3} + I_2(R_2 + R_3) = -5$$

$$I_2\left[-\frac{R_3^2}{R_2 + R_3} + (R_2 + R_3)\right] = 10\frac{R_3}{R_2 + R_3} - 5 \Rightarrow I_2 = \frac{10\frac{R_3}{R_2 + R_3} - 5}{-\frac{R_3^2}{R_2 + R_3} + (R_2 + R_3)} = -0.000375 = -375\,\mu A$$

$$I_1 = \frac{I_2 R_3 + 10}{R_2 + R_3} = 875\,\mu A$$

Hand calculations for A

Part 2

A.

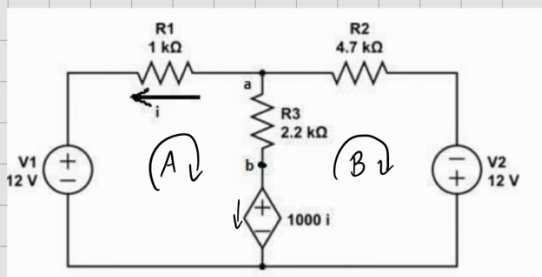


Figure 3.2

$i = -I_A$

KVL A

$-12 + I_A 1K + (I_A - I_B) 2.2K + 1K\, i = 0$

$I_A 2.2K - I_B 2.2K = 12$

KVL B:

$-1K\, i + (I_B - I_A) 2.2K + I_B 4.7K - 12 = 0$

$-I_A 1.2K + I_B 6.9K = 12$

Solve using matrix operations where col 1 is I_A and col 2 is I_B

$$\begin{bmatrix} 2200 & -2200 & 12 \\ -1200 & 6900 & 12 \end{bmatrix} \overset{\text{rref}}{\Rightarrow} \begin{bmatrix} 1 & 0 & 8.708\,\text{mA} \\ 0 & 1 & 3.253\,\text{mA} \end{bmatrix}$$

Because $i = -I_A$, $i = -8.708\,\text{mA}$

B. The voltage $V_{ab} = R_3(I_A - I_B) = 2.2\,\text{k}\Omega(8.708\,\text{mA} - 3.253\,\text{mA}) = 12\,\text{V}$